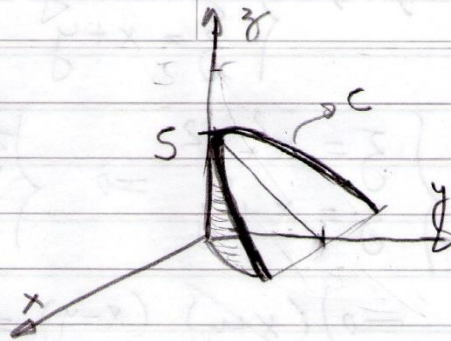


TP4)

1) a)

$$\begin{cases} y = x^2 \\ y + z = 5 \end{cases}$$

$$A = (2, 4, 1)$$



$$\begin{cases} y = x^2 \\ z = 5 - y \end{cases} \Rightarrow \begin{cases} y = x^2 \\ z = 5 - x^2 \end{cases}$$

$$\vec{f}(t) = (t, t^2, 5 - t^2) \quad \vec{f}(t_0) = (2, 4, 1) \Rightarrow t_0 = 2$$

$$\vec{f}'(t) = (1, 2t, -2t) \Rightarrow \vec{f}'(2) = (1, 4, -4)$$

Recta tangente $(x, y, z) = (2, 4, 1) + \lambda(1, 4, -4) \quad \lambda \in \mathbb{R}$

Plano normal

$$[(x, y, z) - (2, 4, 1)] \cdot (1, 4, -4) = 0$$

$$(x-2) + 4(y-4) - 4(z-1) = 0 \Rightarrow x + 4y - 4z = 14$$

Otra forma $\rightarrow$ vector director normal al plano $(1, 4, -4)$

$$Ax + By + Cz + D = 0$$

$$1 \cdot x + 4y - 4z + D = 0 \quad \text{pasa por } (2, 4, 1)$$

$$2 + 4 \cdot 4 - 4 \cdot 1 + D = 0 \Rightarrow D = -14$$

$$x + 4y - 4z - 14 = 0$$

C es plano porque sus pts pertenecen al plano $y + z = 5$.

$$1) b) \begin{cases} z = x^2 - y^2 \\ z = x + y \end{cases} \quad A = (2, 1, 3)$$

$$\begin{cases} z = x^2 - y^2 \\ z = x + y \end{cases} = 0 \quad \begin{cases} z = (x+y)(x-y) \\ z = x + y \end{cases} = 0$$

$$= 0 \Rightarrow \begin{cases} (x+y)(x-y) = x+y \\ z = x+y \end{cases} = 0 \Rightarrow \begin{cases} x-y = 1 \\ z = x+y \end{cases} \Rightarrow$$

$$= 0 \Rightarrow \begin{cases} y = x-1 \\ z = x+x-1 \end{cases} \Rightarrow \begin{cases} y = x-1 \\ z = 2x-1 \end{cases}$$

$$\vec{r}(t) = (t, t-1, 2t-1) \quad \vec{r}(t_0) = (2, 1, 3) \Rightarrow t_0 = 2$$

$$\vec{r}'(t) = (1, 1, 2) \Rightarrow \vec{r}'(t_0) = (1, 1, 2)$$

Recta tangente $(x, y, z) = (2, 1, 3) + \lambda(1, 1, 2) \quad \lambda \in \mathbb{R}$

Plano normal $[(x, y, z) - (2, 1, 3)] \cdot (1, 1, 2) = 0 \Rightarrow$

$$\Rightarrow x-2 + y-1 + 2(z-3) = 0 \Rightarrow$$

$$\Rightarrow x + y + 2z = 9$$

es una curva plana : sus puntos pertenecen al plano $z = x + y$

$$1) c) \begin{cases} x^2 + y^2 + z^2 = 8 \\ z = \sqrt{x^2 + y^2} \end{cases} \Rightarrow \begin{cases} x^2 + y^2 + x^2 + y^2 = 8 \\ z = \sqrt{x^2 + y^2} \end{cases} \Rightarrow 0$$

$$\Rightarrow \begin{cases} x^2 + y^2 = 4 \\ z = \sqrt{x^2 + y^2} \end{cases} \rightarrow \begin{cases} x = 2 \cos t \\ y = 2 \sin t \end{cases} \text{ Luego:}$$

$$z = \sqrt{4 \cos^2 t + 4 \sin^2 t} \Rightarrow z = 2$$

$$\vec{r}(t) = (2 \cos t, 2 \sin t, 2) \quad \vec{r}(t_0) = (0, 2, 2) \Rightarrow 0$$

$$\Rightarrow \begin{cases} 2 \cos t_0 = 0 \\ 2 \sin t_0 = 2 \\ 2 = 2 \end{cases} \Rightarrow t_0 = \pi/2$$

$$\vec{r}'(t) = (-2 \sin t, 2 \cos t, 0) \Rightarrow \vec{r}'(\pi/2) = (-2, 0, 0)$$

Recta tangente $(x, y, z) = (0, 2, 2) + \lambda (-2, 0, 0) \quad \lambda \in \mathbb{R}$

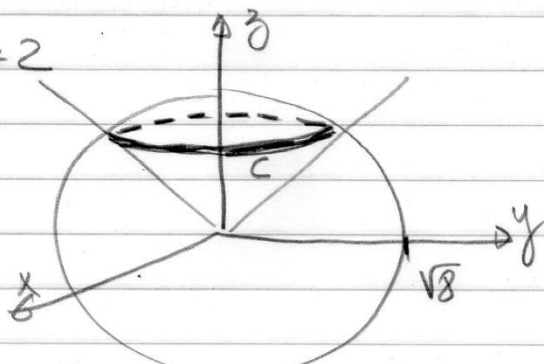
Plano normal

$$[(x, y, z) - (0, 2, 2)] \cdot (-2, 0, 0) = 0$$

$$-2x = 0 \Rightarrow x = 0 \quad \text{plano } (yz)$$

Es una curva plana: sus puntos pertenecen

al plano $z = 2$



$$2) \quad \bar{X} = (u^2, u-2, u+3) \quad \bar{X}(3) = (9, 1, 6)$$

$$\bar{X}' = (2u, 1, 1) \quad \bar{X}'(3) = (6, 1, 1)$$

Recta teste

$$(x, y, z) = \lambda(6, 1, 1) + (9, 1, 6) \quad \lambda \in \mathbb{R} \rightarrow \begin{cases} x = 6\lambda + 9 \\ y = \lambda + 1 \\ z = \lambda + 6 \end{cases}$$

a) eje $z \rightarrow x=0 \wedge y=0 \Rightarrow$

$$\Rightarrow \begin{cases} 6\lambda + 9 = 0 \\ \lambda + 1 = 0 \end{cases} \Rightarrow \begin{cases} \lambda = -1 \\ 6 \cdot (-1) + 9 = 0 \end{cases} \Rightarrow \begin{matrix} \lambda = -1 \\ z = 0 \end{matrix} \quad \underline{\underline{\text{abs}}}$$

(NO) interseca al eje z

b) $z = x - 2y^2$

$$\lambda + 6 = 6\lambda + 9 - 2(\lambda + 1)^2$$

$$\lambda + 6 = 6\lambda + 9 - 2\lambda^2 - 4\lambda - 2$$

$$2\lambda^2 - \lambda - 1 = 0 \Rightarrow \lambda = \frac{1 \pm \sqrt{1+8}}{4} = \frac{1 \pm 3}{4} \begin{matrix} 1 \\ -\frac{1}{2} \end{matrix}$$

Interseca dos veces:

$$\lambda = 1 \rightarrow (x_0, y_0, z_0) = (15, 2, 7)$$

$$\lambda = -\frac{1}{2} \rightarrow (x_0, y_0, z_0) = (6, \frac{1}{2}, \frac{11}{2})$$

c) $\bar{X} = (r, 2r, \frac{32}{r})$

① $9+6\lambda=r$

de ① y ② $\rightarrow 2(9+6\lambda)=1+\lambda \Rightarrow$

② $\lambda+1=2r$

$\Rightarrow 18+12\lambda=1+\lambda \rightarrow 11\lambda=-17 \rightarrow \lambda=-17/11$

③ $\lambda+6 = \frac{32}{r}$

Verifico si satisface ③:

$$6 - \frac{17}{11} = \frac{32}{9+6\lambda} \Rightarrow 6 - \frac{17}{11} = \frac{32}{9+6(-17/11)} \Rightarrow$$

$$\Rightarrow \frac{49}{11} = -\frac{352}{3} \quad \text{absurdo}$$

(NO) interseca

03)

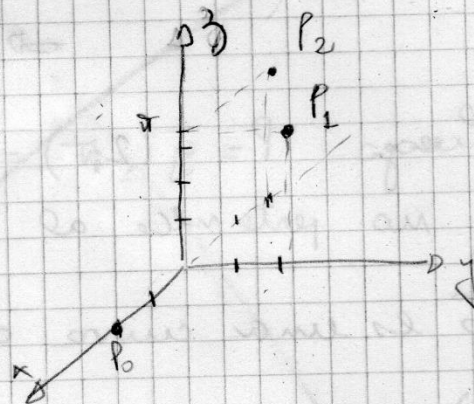
$$\begin{cases} x = 2 \cos t \\ y = 2 \sin t \\ z = t \end{cases}$$

$$0 \leq t \leq 2\pi$$

$$t=0 \Rightarrow P_0 = (2, 0, 0)$$

$$t = \frac{\pi}{2} \Rightarrow P_1 = (0, 2, \frac{\pi}{2})$$

$$t = \pi \Rightarrow P_2 = (-2, 0, \pi)$$



$$\vec{v} = P_1 - P_0 = (0, 2, \frac{\pi}{2}) - (2, 0, 0) = (-2, 2, \frac{\pi}{2})$$

$$\vec{w} = P_2 - P_0 = (-2, 0, \pi) - (2, 0, 0) = (-4, 0, \pi)$$

$$(x, y, z) = (2, 0, 0) + \lambda(-2, 2, \frac{\pi}{2}) + \mu(-4, 0, \pi) \quad \begin{matrix} \lambda \in \mathbb{R} \\ \mu \in \mathbb{R} \end{matrix}$$

$$\begin{cases} x = 2 - 2\lambda - 4\mu \\ y = 2\lambda \\ z = \frac{\pi}{2}\lambda + \mu\pi \end{cases} \Rightarrow \begin{cases} \lambda = \frac{y}{2} \\ x = 2 - y - 4\mu \\ z = \frac{\pi}{2} \frac{y}{2} + \mu\pi \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \lambda = \frac{y}{2} \\ \mu = \frac{x+y-2}{-4} \\ z = \frac{\pi}{2} \frac{y}{2} + \frac{x+y-2}{-4} \pi \end{cases}$$

$$z = \left(\frac{\pi}{4} - \frac{\pi}{4}\right)y = \frac{\pi}{4}x + \frac{\pi}{2} \Rightarrow \boxed{z = -\frac{\pi}{4}x + \frac{\pi}{2}}$$

Verifico que los pts pertenezcan al plano:

$$\bullet (2, 0, 0) \rightarrow 0 = -\frac{\pi}{4} \cdot 2 + \frac{\pi}{2} = 0 \quad 0 = 0 \quad \underline{\text{verif}}$$

$$\bullet (0, 2, \frac{\pi}{2}) \rightarrow \frac{\pi}{2} = -\frac{\pi}{4} \cdot 0 + \frac{\pi}{2} = 0 \quad \frac{\pi}{2} = \frac{\pi}{2} \quad \underline{\text{verif}}$$

$$\bullet (-2, 0, \pi) \rightarrow \pi = -\frac{\pi}{4} (-2) + \frac{\pi}{2} = 0 \quad \pi = \pi \quad \underline{\text{verif}}$$

Si $\mathcal{C}$ es plano $\Rightarrow \forall P: P \in \mathcal{C} \Rightarrow P \in \text{pl} (z = -\frac{\pi}{4}x + \frac{\pi}{2})$

$$\text{Sea } P = \bar{g}(2\pi) = (2, 0, 2\pi)$$

$$\text{Verifico si } P \in \text{plano} : 2\pi = -\frac{\pi}{4} \cdot 2 + \frac{\pi}{2} \Rightarrow$$

$$\Rightarrow 2\pi = 0 \quad \underline{\text{absurdo}}$$

Luego $P = \bar{g}(2\pi) = (2, 0, 2\pi)$ pertenece a la curva
y no pertenece al plano.

$\mathcal{C}$ es una curva alaredada

$$4) a) \begin{aligned} f'_x &= 4x^3 + 2y + y^3 \\ f'_y &= 2x + 3xy^2 \end{aligned}$$

$$b) \begin{aligned} f'_x &= 2ye^{2x} \\ f'_y &= e^{2x} + 3ze^{3y} \\ f'_z &= e^{3y} \end{aligned}$$

$$c) \begin{aligned} f'_x &= e^{x^2+y^2} + 2x^2 e^{x^2+y^2} \\ f'_y &= 2yx e^{x^2+y^2} \end{aligned}$$

$$d) f'_x = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2}$$

$$f'_y = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

$$e) \begin{aligned} f'_x &= -e^{\sin x} \\ f'_y &= e^{\sin y} \end{aligned}$$

$$f) \begin{aligned} f'_x &= e^{(x^2+y^2)^2} \cdot 2x - e^{x^2} \\ f'_y &= e^{(x^2+y^2)^2} \cdot 2y \end{aligned}$$

$$\begin{aligned}
 5) a) \frac{\partial f}{\partial x}(1,2) &= \lim_{h \rightarrow 0} \frac{f(1+h, 2) - f(1, 2)}{h} = \\
 &= \lim_{h \rightarrow 0} \frac{3(1+h)^2 + 2(1+h) \cdot 2 - (3 \cdot 1^2 + 2 \cdot 1 \cdot 2)}{h} = \\
 &= \lim_{h \rightarrow 0} \frac{3(1+2h+h^2) + \cancel{4} + 4h - \cancel{3} - \cancel{4}}{h} = \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{3} + 6h + 3h^2 + 4h - \cancel{3}}{h} = \lim_{h \rightarrow 0} \frac{\cancel{h}(10+3h)}{\cancel{h}} = 10
 \end{aligned}$$

$$\frac{\partial f}{\partial x} = 6x + 2y \Rightarrow \frac{\partial f}{\partial x}(1,2) = 6 \cdot 1 + 2 \cdot 2 = 10$$

$$\begin{aligned}
 \frac{\partial f}{\partial y}(1,2) &= \lim_{k \rightarrow 0} \frac{f(1, 2+k) - f(1, 2)}{k} = \\
 &= \lim_{k \rightarrow 0} \frac{3 \cdot 1^2 + 2 \cdot 1(2+k) - (3 \cdot 1^2 + 2 \cdot 1 \cdot 2)}{k} = \\
 &= \lim_{k \rightarrow 0} \frac{\cancel{3} + \cancel{4} + 2k - \cancel{3} - \cancel{4}}{k} = \lim_{k \rightarrow 0} \frac{2k}{k} = 2
 \end{aligned}$$

$$\frac{\partial f}{\partial y} = 2x \Rightarrow \frac{\partial f}{\partial y}(1,2) = 2 \cdot 1 = 2$$

$$5) b) f(x,y) = \begin{cases} x^2 + 2y & \text{si } x \geq 1 \\ 2x - y & \text{si } x < 1 \end{cases} \quad A = (1,1)$$

La función está partida en $A = (1,1)$. Debemos evaluar las derivadas por definición.

$$f'_x(1,1) = \lim_{h \rightarrow 0} \frac{f[(1,1) + h(1,0)] - \overbrace{f(1,1)}^{=3}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(1+h, 1) - 3}{h}$$

$$\bullet \lim_{h \rightarrow 0^+} \frac{f(1+h, 1) - 3}{h} = \lim_{h \rightarrow 0^+} \frac{(1+h)^2 + 2 - 3}{h} =$$

(primera componente mayor que 1)

$$= \lim_{h \rightarrow 0^+} \frac{1 + 2h + h^2 - 1}{h} = \lim_{h \rightarrow 0^+} \frac{h(2+h)}{h} = 2$$

$$\bullet \lim_{h \rightarrow 0^-} \frac{f(1+h, 1) - 3}{h} = \lim_{h \rightarrow 0^-} \frac{2(1+h) - 1 - 3}{h} =$$

(primera componente menor que 1)

$$= \lim_{h \rightarrow 0^-} \frac{2 + 2h - 4}{h} = \lim_{h \rightarrow 0^-} \frac{2h - 2}{h} = \infty$$

$\nexists$ límite $\Rightarrow \nexists f'_x(1,1)$

$$f'_y(1,1) = \lim_{h \rightarrow 0} \frac{f[(1,1) + h(0,1)] - \overbrace{f(1,1)}^{=3}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(1, 1+h) - 3}{h} = \lim_{h \rightarrow 0} \frac{1^2 + 2(1+h) - 3}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{1 + 2 + 2h - 3}{h} = (2) = f'_y(1,1)$$

$$5)c) f(x,y) = \sqrt{x^4 + 2y^2} \quad A = (0,0)$$

$$f'_x = \frac{1}{2\sqrt{x^4 + 2y^2}} \cdot 4x^3$$

$$f'_y = \frac{1}{2\sqrt{x^4 + 2y^2}} \cdot 4y$$

no pueden
evaluarse en (0,0)

Aplicamos la definición:

$$f'_x(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(1,0)] - \overbrace{f(0,0)}^{=0}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(h,0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h^4}}{h} = \lim_{h \rightarrow 0} \frac{h^2}{h} = 0$$

Luego $f'_x(0,0) = 0$

$$f'_y(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(0,1)] - \overbrace{f(0,0)}^{=0}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(0,h)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{2h^2}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{2}|h|}{h}$$

$$\text{Si } h \rightarrow 0^+ : \lim_{h \rightarrow 0^+} \frac{\sqrt{2}h}{h} = \sqrt{2}$$

$$\text{Si } h \rightarrow 0^- : \lim_{h \rightarrow 0^-} \frac{\sqrt{2}(-h)}{h} = -\sqrt{2}$$

$\nexists$ límite

Luego $\nexists f'_y(0,0)$

5)d) No puede verificarse por regla de derivación porque la función módulo no es derivable en 0

$$f'_x(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0)+h(1,0)] - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(h,0)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

$$\otimes \left\{ \begin{array}{l} \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \\ \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1 \end{array} \right\} \not\equiv \neq f'_x(0,0)$$

$$f'_y(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0)+h(0,1)] - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(0,h)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} \quad \text{por } \otimes \text{ o } \neq f'_y(0,0)$$

$$6) f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$$

a) Continuidad

i) $\exists f(0,0) = 0$

ii) $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$

$$\boxed{y = mx} \quad \lim_{x \rightarrow 0} \frac{x \cdot mx}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2(1+m^2)} =$$

$$= \frac{m}{1+m^2} \quad \text{depende de } m \therefore \nexists L \Rightarrow \text{no es continua en } \vec{0}$$

b) Derivabilidad $\vec{r} = (a,b) / a^2 + b^2 = 1$

$$f'((0,0), \vec{r}) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(a,b)] - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(ha, hb)}{h} = \lim_{h \rightarrow 0} \frac{\frac{ha \cdot hb}{h^2 a^2 + h^2 b^2}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h^2} ab}{\cancel{h^2} (a^2 + b^2)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{ab}{h}$$

$\underbrace{a^2 + b^2}_{=1}$

si $ab \neq 0$ $\lim_{h \rightarrow 0} \frac{ab}{h} = \infty \therefore \nexists f'(\vec{0}, \vec{r})$

si $ab = 0$ $\lim_{h \rightarrow 0} \frac{ab}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$

si $ab = 0 \quad \exists f'(\vec{0}, \vec{r}) = 0$

$ab = 0 \Rightarrow a = 0 \vee b = 0$

$a = 0$	$\vec{r}_1 = (0, 1) \quad \vec{r}_2 = (0, -1)$
$b = 0$	$\vec{r}_3 = (1, 0) \quad \vec{r}_4 = (-1, 0)$

$$7) \quad g(x,y) = \begin{cases} \frac{x^3 y}{x^6 + y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Derivabilidad

$$\vec{r} = (a,b) / a^2 + b^2 = 1$$

$$g'(\vec{0}, \vec{r}) = \frac{\partial g}{\partial \vec{r}}(0,0) = \lim_{h \rightarrow 0} \frac{g[(0,0) + h(a,b)] - g(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{g(ha, hb)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^3 a^3 h b}{h^6 a^6 + h^2 b^2}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{h^4 a^3 b}{h^2 (h^4 a^6 + b^2)} \cdot \frac{1}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{h a^3 b}{h^4 a^6 + b^2} = 0$$

Luego g es derivable en toda dirección y sentido en $(0,0)$

Continuidad

$$i) \quad g(0,0) = 0$$

$$ii) \quad \lim_{(x,y) \rightarrow (0,0)} g(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$$

$$l_{12} = l_{21} = 0$$

$$\boxed{y = x^3} \quad \lim_{x \rightarrow 0} \frac{x^3 x^3}{x^6 + x^6} = \frac{1}{2} \neq l_{12} = 0 \quad \text{AL} \Rightarrow g \text{ no es continua en } \vec{0}$$

g es derivable en toda dirección y sentido en $(0,0)$ pero no es continua en $(0,0)$

$$8) f(x,y) = \sqrt[3]{x} \sqrt[3]{y}$$

a) Derivabilidad en $(0,0)$ $\vec{v} = (a,b) / a^2 + b^2 = 1$

$$\frac{\partial f}{\partial \vec{v}}(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(a,b)] - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(ha, hb)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt[3]{ha} \sqrt[3]{hb}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{h^{2/3} (ab)^{1/3}}{h} = \lim_{h \rightarrow 0} \frac{(ab)^{1/3}}{\sqrt[3]{h}}$$

Si $ab \neq 0$: $\lim_{h \rightarrow 0} \frac{(ab)^{1/3}}{\sqrt[3]{h}} = \infty \therefore \nexists$ deriv. direccional

Si $ab = 0$: $\lim_{h \rightarrow 0} \frac{0}{\sqrt[3]{h}} = 0$

$ab = 0 \rightarrow \begin{cases} a = 0 & \vec{v}_1 = (0,1) & \vec{v}_2 = (0,-1) \\ b = 0 & \vec{v}_3 = (1,0) & \vec{v}_4 = (-1,0) \end{cases}$

Como $\vec{v}_1 = \vec{e}_1 \Rightarrow \frac{\partial f}{\partial \vec{e}_1}(0,0) = f'_x(0,0) = 0$

Como $\vec{v}_3 = \vec{e}_2 \Rightarrow \frac{\partial f}{\partial \vec{e}_2}(0,0) = f'_y(0,0) = 0$

b) Continuidad

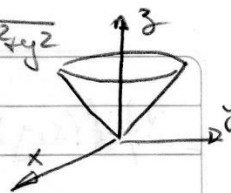
i) $f(0,0) = 0$ ii) $\lim_{(x,y) \rightarrow (0,0)} \sqrt[3]{x} \sqrt[3]{y} = 0 = L$

iii) $f(0,0) = L$

Luego f es continua en $(0,0)$

9) $f(x,y) = \sqrt{4x^2 + y^2}$

gráfica: $z = \sqrt{4x^2 + y^2}$



Continuidad i) $f(0,0) = 0$

ii) $\lim_{(x,y) \rightarrow (0,0)} \sqrt{4x^2 + y^2} = 0 = L$

iii) $f(0,0) = L$

Luego f es continua en $(0,0)$

Derivadas parciales

$$f'_x = \frac{1}{2\sqrt{4x^2 + y^2}} \cdot 8x$$

$$f'_y = \frac{1}{2\sqrt{4x^2 + y^2}} \cdot 2y$$

no pueden
evaluarse
en $(0,0)$

Por definición:

$$f'_x(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(1,0)] - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(h,0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{4h^2}}{h} = \lim_{h \rightarrow 0} \frac{2|h|}{h}$$

$$\lim_{h \rightarrow 0^+} \frac{2h}{h} = 2$$

$$\lim_{h \rightarrow 0^-} \frac{2(-h)}{h} = -2$$

$\nexists \lim_{h \rightarrow 0} \frac{f'_x(0,0)}$

$$f'_y(0,0) = \lim_{h \rightarrow 0} \frac{f[(0,0) + h(0,1)] - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{f(0,h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h^2}}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} \rightarrow \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

$$\lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

$\nexists \lim_{h \rightarrow 0} \frac{f'_y(0,0)}$

$$10) \quad \vec{r} = (3, 4) \Rightarrow \vec{r} = \frac{(3, 4)}{\sqrt{9+16}} = \left(\frac{3}{5}, \frac{4}{5}\right)$$

$$\frac{\partial f}{\partial \vec{r}}(1, -1) = \lim_{h \rightarrow 0} \frac{f\left[\left(1, -1\right) + h\left(\frac{3}{5}, \frac{4}{5}\right)\right] - f(1, -1)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{3}{5}h, -1 + \frac{4}{5}h\right) - (1 - 1 + 2)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{1 - \left(1 + \frac{3}{5}h\right)^2 + 2\left(-1 + \frac{4}{5}h\right)^2 - 2}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{1} - \cancel{1} - \frac{9}{25}h^2 - \frac{6}{5}h + \cancel{2} + \frac{32}{25}h^2 - \frac{16}{5}h - \cancel{2}}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{\frac{23}{25}h^2 - \frac{22}{5}h}{h} = \lim_{h \rightarrow 0} \frac{\frac{23}{25}h - \frac{22}{5}}{1} =$$

$$= \boxed{-22/5}$$

$$11) a) \quad f(x,y) = \begin{cases} \frac{xy-x}{x^2+(y-1)^2} & \text{if } (x,y) \neq (0,1) \\ 0 & \text{if } (x,y) = (0,1) \end{cases}$$

Derivability test $\vec{v} = (a,b) / a^2 + b^2 = 1$

$$\frac{\partial f}{\partial \vec{v}}(0,1) = \lim_{h \rightarrow 0} \frac{f[(0,1) + h(a,b)] - f(0,1)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(ha, 1+hb)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{ha(1+hb) - ha}{h^2a^2 + (1+hb-1)^2} =$$

$$= \lim_{h \rightarrow 0} \frac{ha + h^2ab - ha}{h^2a^2 + h^2b^2} \cdot \frac{1}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{h^2ab}{h^3(a^2+b^2)} = \lim_{h \rightarrow 0} \frac{ab}{h}$$

$\underbrace{a^2+b^2}_{=1}$

If $ab \neq 0$ $\lim_{h \rightarrow 0} \frac{ab}{h} = \infty$ $\nexists$ no is derivable

If $ab = 0$ $\lim_{h \rightarrow 0} \frac{ab}{h} = 0$

$ab = 0 \rightarrow a = 0 \quad \vec{v}_1 = (0,1) \quad \vec{v}_2 = (0,-1)$
 $\rightarrow b = 0 \quad \vec{v}_3 = (1,0) \quad \vec{v}_4 = (-1,0)$

$$11) b) f(x,y) = \begin{cases} \frac{y^2}{x} & \text{si } (x,y) \neq (0,y) \rightarrow x \neq 0 \\ 0 & \text{si } (x,y) = (0,y) \rightarrow x = 0 \end{cases}$$

Derivabilidad $\vec{r} = (a,b) / a^2 + b^2 = 1$

$$\frac{\partial f}{\partial \vec{r}}(0,0) = \lim_{h \rightarrow 0} \frac{f(\overbrace{(0,0) + h(a,b)}^{=0}) - f(0,0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{f(ha, hb)}{h} = (*)$$

• Si $ha = 0 \Rightarrow a = 0$ (para $(0,1)$ ó $(0,-1)$)

$$(*) = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

• Si $ha \neq 0 \Rightarrow a \neq 0$

$$(*) = \lim_{h \rightarrow 0} \frac{\frac{h^2 b^2}{ha}}{h} = \lim_{h \rightarrow 0} \frac{h^2 b^2}{h^2 a} = \frac{b^2}{a}$$

Es derivable en toda dirección y sentido en $(0,0)$ -

• Para $\vec{r}_1 = (0,1)$ y $\vec{r}_2 = (0,-1)$ la derivada es nula

• Para cualquier otro vector $\vec{r} = (a,b)$, la derivada es $\frac{b^2}{a}$

Por ejemplo: hallamos $f'_x(0,0) = \frac{\partial f}{\partial (1,0)}(0,0) = \frac{0^2}{1} = 0$

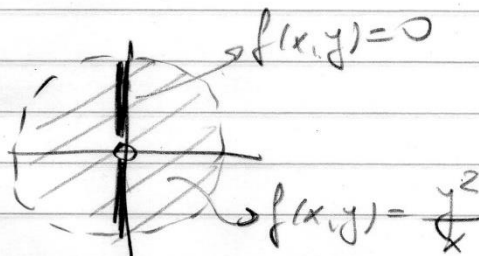
Hallamos $f'_y(0,0) = 0$ (corresponde a $\vec{r}_1$)

Hallamos $\frac{\partial f}{\partial (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})}(0,0) = \frac{\left(\frac{1}{\sqrt{2}}\right)^2}{\frac{1}{\sqrt{2}}} = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{\sqrt{2}}{2}$

Realizamos continuidad en $(0,0)$

i) $f(0,0) = 0$

ii) $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$



$\boxed{x=0}$ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 0 = 0$

$\boxed{x \neq 0}$ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x}$

$\boxed{x=y^2}$ $\lim_{y \rightarrow 0} \frac{y^2}{y^2} = 1 \neq \lim_{x=0} = 0 \neq L=0$

$\Rightarrow f$ no es continua en $(0,0)$

f es derivable en toda dirección y

sentido en $(0,0)$, pero no es continua en $(0,0)$

12)

$$a) f(x, y) = \ln(x^2 + y^2)$$

$$\text{Dom } f = \mathbb{R}^2 - \{(0, 0)\}$$

$$\frac{\partial f}{\partial x} = \frac{2x}{x^2 + y^2}$$

$$\frac{\partial f}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$\text{Dom } \frac{\partial f}{\partial x} = \text{Dom } \frac{\partial f}{\partial y} = \mathbb{R}^2 - \{(0, 0)\}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(x^2 + y^2) - 2x \cdot 2x}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2 + y^2) - 2y \cdot 2y}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{0(x^2 + y^2) - 2y \cdot 2x}{(x^2 + y^2)^2}$$

$$\text{Dom } \frac{\partial^2 f}{\partial x^2} = \text{Dom } \frac{\partial^2 f}{\partial y^2} = \text{Dom } \frac{\partial^2 f}{\partial x \partial y} = \mathbb{R}^2 - \{(0, 0)\}$$

$$12)b) \quad \bar{f}(x,y) = \left(x \ln y, \frac{y}{x} \right)$$

$$\text{Dom } \bar{f} = \{ (x,y) \in \mathbb{R}^2 \mid y > 0 \wedge x \neq 0 \} = A$$

$$\frac{\partial \bar{f}}{\partial x} = \left(\ln y, -\frac{y}{x^2} \right)$$

$$\frac{\partial \bar{f}}{\partial y} = \left(\frac{x}{y}, \frac{1}{x} \right)$$

$$\text{Dom } \frac{\partial \bar{f}}{\partial x} = \text{Dom } \frac{\partial \bar{f}}{\partial y} = A$$

$$\frac{\partial^2 \bar{f}}{\partial x^2} = \left(0, \frac{2y}{x^3} \right)$$

$$\frac{\partial^2 \bar{f}}{\partial y^2} = \left(-\frac{x}{y^2}, 0 \right)$$

$$\frac{\partial^2 \bar{f}}{\partial x \partial y} = \frac{\partial^2 \bar{f}}{\partial y \partial x} = \left(\frac{1}{y}, -\frac{1}{x^2} \right)$$

$$\text{Dom } \frac{\partial^2 \bar{f}}{\partial x^2} = \text{Dom } \frac{\partial^2 \bar{f}}{\partial y^2} = \text{Dom } \frac{\partial^2 \bar{f}}{\partial x \partial y} = A$$

$$12)c) \quad f(x,y) = \begin{cases} \frac{x^2}{x^2+y^2} & \text{si } \bar{x} \neq \bar{0} \\ 0 & \text{si } \bar{x} = \bar{0} \end{cases}$$

$$\text{Dom } f = \mathbb{R}^2$$

$$\frac{\partial f}{\partial x} \Big|_{(x,y) \neq (0,0)} = \frac{2x(x^2+y^2) - 2x \cdot x^2}{(x^2+y^2)^2} = \frac{2xy^2}{(x^2+y^2)^2}$$

$$\begin{aligned} \frac{\partial f}{\partial x}(0,0) &= \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^2}{h^2} - 0}{h} = \\ &= \lim_{h \rightarrow 0} \frac{1}{h} = \infty \neq \frac{\partial f}{\partial x}(0,0) \end{aligned}$$

$$\left. \frac{\partial f}{\partial y} \right|_{(x,y) \neq (0,0)}$$

$$\frac{\partial f}{\partial y} = \frac{0 - 2yx^2}{(x^2+y^2)^2} = \frac{-2yx^2}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

$$\text{Dom } \frac{\partial f}{\partial x} = \mathbb{R}^2 - \{(0,0)\}$$

$$\text{Dom } \frac{\partial f}{\partial y} = \mathbb{R}^2$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{2y^2(x^2+y^2)^2 - 2(x^2+y^2) \cdot 2x \cdot 2xy^2}{(x^2+y^2)^4}$$

$$\left. \frac{\partial^2 f}{\partial y^2} \right|_{(x,y) \neq (0,0)} = \frac{-2x^2(x^2+y^2)^2 - 2(x^2+y^2) \cdot 2y \cdot (-2yx^2)}{(x^2+y^2)^4}$$

$$\begin{aligned} \frac{\partial^2 f}{\partial y^2}(0,0) &= \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial y}(0,h) - \frac{\partial f}{\partial y}(0,0)}{h} = \\ &= \lim_{h \rightarrow 0} \frac{0-0}{h} = 0 \end{aligned}$$

$$(x,y) \neq (0,0) \rightarrow \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{-4xy(x^2+y^2)^2 - 2(x^2+y^2) \cdot 2x \cdot (-2yx^2)}{(x^2+y^2)^4}$$

$$\cancel{\frac{\partial f}{\partial x}(0,0)} \Rightarrow \cancel{\frac{\partial^2 f}{\partial y \partial x}(0,0)}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial y}(h,0) - \frac{\partial f}{\partial y}(0,0)}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = 0$$

$$\text{Dom } \frac{\partial^2 f}{\partial y^2} = \text{Dom } \frac{\partial^2 f}{\partial x \partial y} = \mathbb{R}^2$$

$$\text{Dom } \frac{\partial^2 f}{\partial x^2} = \text{Dom } \frac{\partial^2 f}{\partial y \partial x} = \mathbb{R}^2 - \{(0,0)\}$$

13)

$$x(t) = 3t^2$$

$$y(t) = 2-t$$

$$z(t) = 2t^2$$

$$\text{Sup } S \rightarrow x = y^2 + z = 0 \quad 3t^2 = (2-t)^2 + 2t^2 \Rightarrow$$

$$\Rightarrow 3t^2 = 4 - 4t + t^2 + 2t^2 \Rightarrow 4t = 4 \Rightarrow t = 1$$

$$\bar{x}'(t) = (6t, -1, 4t)$$

$$\text{veloc. } \bar{v} = \bar{x}'(1) = (6, -1, 4)$$

$$\bar{x}''(t) = (6, 0, 4)$$

$$\text{acelerac. } \bar{a} = \bar{x}''(1) = (6, 0, 4)$$

NOTA

El ángulo θ entre $\bar{v}$ y $\bar{a}$ es: $\theta = \arccos \left| \frac{\bar{v} \cdot \bar{a}}{\|\bar{v}\| \|\bar{a}\|} \right|$

$$\Rightarrow \theta = \frac{\arccos \frac{(6, -1, 4) \cdot (6, 0, 4)}{\sqrt{36+1+16} \sqrt{36+0+16}}}{\sqrt{36+1+16} \sqrt{36+16}} = \frac{\arccos \frac{36+16}{\sqrt{53} \sqrt{52}}}{\sqrt{53} \sqrt{52}} = \boxed{\arccos \frac{\sqrt{521}}{\sqrt{53}} = \alpha}$$

$$\text{Plano } \pi \rightarrow x - z + 2y = 7 \Rightarrow 3t^2 - 2t^2 + 2(2-t) = 7 \Rightarrow$$

$$\Rightarrow t^2 + 4 - 2t = 7 \Rightarrow t^2 - 2t - 3 = 0 \Rightarrow$$

$$\Rightarrow t = \frac{2 \pm \sqrt{4+12}}{2} \Rightarrow t = \frac{2 \pm 4}{2} \Rightarrow t_1 = 3 \vee t_2 = -1$$

$t_2 = -1$ absurdo porque $t \geq 0$ por hipótesis
 $t = 3$ es el instante en que el móvil atraviesa el plano π .

$t = 1 \rightarrow$ atraviesa S
 $t = 3 \rightarrow$ atraviesa π } $\Rightarrow$ tarda 2 segundos en llegar de S hasta π

$$14) L'(\bar{A}, \tilde{x}) = \lim_{h \rightarrow 0} \frac{L(\bar{A} + h\tilde{x}) - L(\bar{A})}{h} =$$

$$\stackrel{\text{prop TL}}{=} \lim_{h \rightarrow 0} \frac{L(\bar{A}) + L(h\tilde{x}) - L(\bar{A})}{h} = \lim_{h \rightarrow 0} \frac{L(h\tilde{x})}{h} =$$

$$= L(\tilde{x})$$

$$\text{Luego } L(\bar{A}, \tilde{x}) = L(\tilde{x}) \quad \forall \bar{A}, \tilde{x} \in \mathbb{R}^n$$

15) Verificamos que f es una TL

$$i) f(\bar{x} + \bar{y}) = \bar{k}(\bar{x} + \bar{y}) = \bar{k} \cdot \bar{x} + \bar{k} \cdot \bar{y} = f(\bar{x}) + f(\bar{y})$$

$$ii) f(a\bar{x}) = \bar{k}(a\bar{x}) = a(\bar{k}\bar{x}) = a f(\bar{x})$$

i) y ii) $\Rightarrow f$ es TL

$$f'[(2,1), (3,4)] = f(3,4) = (k_1, k_2) \cdot (3,4) = 3k_1 + 4k_2 = 2$$

$$f'[(3,5), (-1,1)] = f(-1,1) = (k_1, k_2) \cdot (-1,1) = -k_1 + k_2 = 4$$

$$\Rightarrow \begin{cases} k_2 = 4 + k_1 \\ 3k_1 + 4(4 + k_1) = 2 \end{cases} \Rightarrow \begin{cases} k_2 = 4 + k_1 \\ 7k_1 = 2 - 16 \end{cases} \Rightarrow \begin{cases} k_1 = -2 \\ k_2 = 2 \end{cases}$$

NOTA

$$\bar{k} = (-2, 2)$$

$$16) \quad \vec{v}'_x = (g(x) + xg'(x), 0)$$

$$\vec{v}'_y = (-1, 2y)$$

$$\vec{v}'_x \perp \vec{v}'_y = 0 \quad \vec{v}'_x \cdot \vec{v}'_y = 0 \Rightarrow g + xg' = 0 \Rightarrow g = -x \frac{dg}{dx} \Rightarrow$$
$$= 0 \int \frac{dx}{x} = \int \frac{dg}{g} \Rightarrow -\ln kx = \ln g \Rightarrow \ln(kx)^{-1} = \ln g$$

$$\Rightarrow g = \frac{1}{kx}$$

$$\vec{v}(1,1) = (3,1) \Rightarrow \left(1 + 1 \frac{1}{k \cdot 1}, 1\right) = (3,1) \Rightarrow$$

$$\Rightarrow \frac{1}{k} = 3 - 1 \Rightarrow k = \frac{1}{2}$$

$$\boxed{g(x) = \frac{2}{x}}$$

$$17) \quad f(x, y) = y^2 + g(x)$$

$$f'_x = g'(x) = 0 \quad f''_{xx} = g''(x)$$

$$f'_y = 2y = 0 \quad f''_{yy} = 2$$

$$f''_{xx} + f''_{yy} = g''(x) + 2 = 0 \Rightarrow g''(x) = -2$$

Sea $u = g'(x)$, luego: $u' = -2 \Rightarrow \frac{du}{dx} = -2 \Rightarrow$

$$\Rightarrow \int du = \int -2 dx \Rightarrow u = -2x + C$$

Por lo tanto: $g'(x) = -2x + C \Rightarrow \frac{dg}{dx} = -2x + C \Rightarrow$

$$\Rightarrow \int dg = \int (-2x + C) dx \Rightarrow g(x) = -x^2 + Cx + k$$

$$f(x, y) = y^2 - x^2 + Cx + k$$

$$f(0, 0) = k = 0 \Rightarrow k = 0$$

$$f'_x = -2x + C$$

$$f'_x(0, 0) = 1 \Rightarrow C = 1$$

$$f(x, y) = y^2 - x^2 + x$$

$$g(x) = x - x^2$$

$$18) \quad \overline{x}_1 = (t+1, g(t), 1+t^2) \\ \overline{x}_2 = (2t, g'(t), 2t^2)$$

$$\overline{x}_1 \parallel \overline{x}_2 \Rightarrow \overline{x}_1' \parallel \overline{x}_2'$$

$$\overline{x}_1' = (1, g'(t), 2t)$$

$$\overline{x}_2' = (2, g''(t), 4t)$$

$$\overline{x}_1' \parallel \overline{x}_2' \Rightarrow (1, g', 2t) = k(2, g'', 4t) \quad \text{para algùn } k \in \mathbb{R}$$

$$\begin{cases} 1 = 2k \\ g' = k g'' \\ 2t = 4kt \end{cases} \Rightarrow \begin{cases} k = 1/2 \\ g' = k g'' \\ k = 1/2 \end{cases} \quad \text{Luego } g' = \frac{1}{2} g''$$

$$\text{Hacemos } u = g' \text{, luego } u = \frac{1}{2} u' \Rightarrow$$

$$\Rightarrow \frac{du}{dt} = 2u \Rightarrow \int \frac{du}{u} = \int 2 dt \Rightarrow \ln u = 2t \Rightarrow$$

$$\Rightarrow e^{2t} = ku \Rightarrow u = ce^{2t}$$

$$\text{Luego : } g' = ce^{2t} \Rightarrow \frac{dg}{dt} = ce^{2t} \Rightarrow \int dg = \int ce^{2t} dt \Rightarrow$$

$$\Rightarrow g(t) = \frac{ce^{2t}}{2} + b$$

$$\overline{K}_1 = (t+1, \frac{ce^{2t}}{2} + b, 1+t^2)$$

$$\overline{K}_2 = (2t, ce^{2t}, 2t^2)$$

$$t=1 \Rightarrow \overline{K}_1(1) = (2, 2, 2) \Rightarrow (2, \frac{ce^2}{2} + b, 2) = (2, 2, 2) \Rightarrow$$

$$\Rightarrow \frac{ce^2}{2} + b = 2 \quad \textcircled{A}$$

$$t=1 \Rightarrow \overline{K}_2(1) = (2, ce^2, 2) = (2, 2, 2) \Rightarrow$$

$$\Rightarrow ce^2 = 2 \quad \textcircled{B}$$

$$\text{De } \textcircled{B} \rightarrow c = 2e^{-2}$$

$$\text{Reemplazando en } \textcircled{A} \rightarrow b = 2 - \frac{ce^2}{2} \Rightarrow b = 2 - \frac{2e^{-2}e^2}{2} = 1$$

$$g(t) = \frac{2e^{-2}e^{2t}}{2} + 1 = \frac{e^{2t-2} + 1}{1}$$

$$19) \frac{\partial y}{\partial t} = 2(x-at)(-a) + 2(x+at)a$$

$$\frac{\partial^2 y}{\partial t^2} = -2a \cdot (-a) + 2a \cdot a = 4a^2 \quad (1)$$

$$\frac{\partial y}{\partial x} = 2(x-at) + 2(x+at)$$

$$\frac{\partial^2 y}{\partial x^2} = 2 + 2 = 4 \quad (2)$$

$$\text{De (1) y (2)} \rightarrow \frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

$$20) \begin{aligned} z'_x &= \text{senh}(y) \cdot \cos(x) \\ z'_{xx} &= -\text{sen}(x) \text{senh}(y) \quad (1) \end{aligned}$$

$$\begin{aligned} z'_y &= \text{sen } x \cosh y \\ z''_{yy} &= \text{sen } x \text{senh } y \quad (2) \end{aligned}$$

$$z''_{xx} + z''_{yy} = 0 \quad \text{De (1) y (2)}$$

21) dist ao origem de $P(x, y, z) \rightarrow \sqrt{x^2 + y^2 + z^2}$
 $U = \frac{k}{\sqrt{x^2 + y^2 + z^2}} \quad k \in \mathbb{R} \quad \text{Dom } U = \mathbb{R}^3 - \{0\}$

$$U'_x = k \left(-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot 2x \right) = -kx (x^2 + y^2 + z^2)^{-3/2}$$

$$U'_y = -ky (x^2 + y^2 + z^2)^{-3/2}$$

$$U'_z = -kz (x^2 + y^2 + z^2)^{-3/2}$$

$$U''_{xx} = -k (x^2 + y^2 + z^2)^{-3/2} - kx \left(-\frac{3}{2} \right) (x^2 + y^2 + z^2)^{-5/2} \cdot 2x$$

$$= -k (x^2 + y^2 + z^2)^{-3/2} + 3kx^2 (x^2 + y^2 + z^2)^{-5/2}$$

$$U''_{yy} = -k(x^2 + y^2 + z^2)^{-5/2} + 3ky^2(x^2 + y^2 + z^2)^{-7/2}$$

$$U''_{zz} = -k(x^2 + y^2 + z^2)^{-3/2} + 3kz^2(x^2 + y^2 + z^2)^{-5/2}$$

$$\begin{aligned} U''_{xx} + U''_{yy} + U''_{zz} &= -3k(x^2 + y^2 + z^2)^{-3/2} + \\ &+ 3k(x^2 + y^2 + z^2)^{-5/2}(x^2 + y^2 + z^2) = \\ &= -3k(x^2 + y^2 + z^2)^{-3/2} + 3k(x^2 + y^2 + z^2)^{-3/2} = 0 \end{aligned}$$

$$22) \vec{f}(x, y, z) = \frac{k(x, y, z)}{(\sqrt{x^2 + y^2 + z^2})^3} = \frac{k(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

Don't $\vec{f} = R^3 - \{0\}$

$$P = \frac{kx}{(x^2 + y^2 + z^2)^{3/2}} \Rightarrow P'_x = \frac{k(x^2 + y^2 + z^2)^{3/2} - \frac{3}{2}(x^2 + y^2 + z^2)^{1/2} kx \cdot 2x}{(x^2 + y^2 + z^2)^3}$$

$$P'_x = k(x^2 + y^2 + z^2)^{-3/2} - 3kx^2(x^2 + y^2 + z^2)^{-5/2}$$

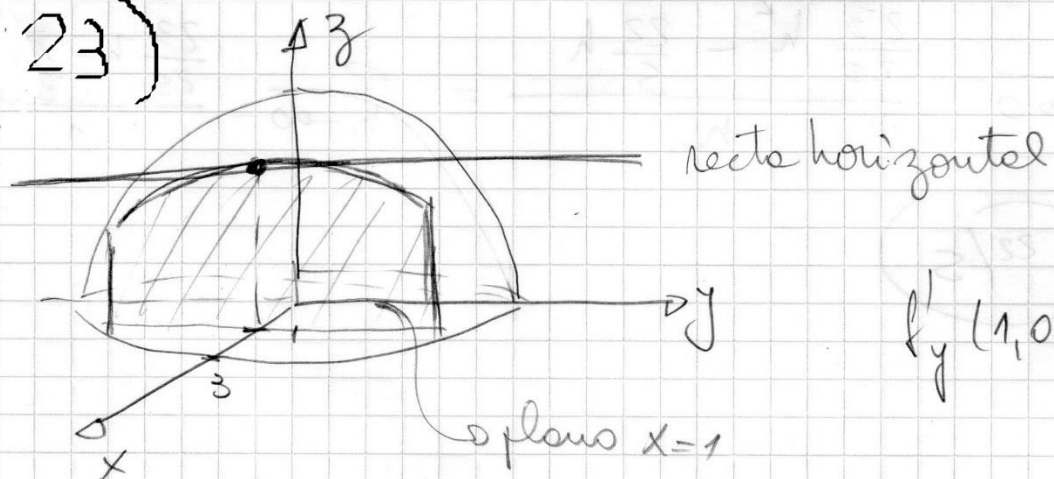
$$Q'_y = k(x^2 + y^2 + z^2)^{-3/2} - 3ky^2(x^2 + y^2 + z^2)^{-5/2}$$

$$R'_z = k(x^2 + y^2 + z^2)^{-3/2} - 3kz^2(x^2 + y^2 + z^2)^{-5/2}$$

$$P'_x + Q'_y + R'_z = 3k(x^2 + y^2 + z^2)^{-3/2} - 3k(x^2 + y^2 + z^2)^{-5/2} \cdot$$

$$\cdot (x^2 + y^2 + z^2) = 3k(x^2 + y^2 + z^2)^{-3/2} - 3k(x^2 + y^2 + z^2)^{-3/2} = 0$$

23)



$$f'_y(1,0) = 0$$